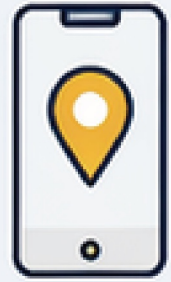


Verification Outcomes Under Representative Scenarios



This stage filters invalid cross-round behaviors before secure aggregation.



Submitted Update

Client update with trajectory, proofs, and metadata



Verification Checks

Run cryptographic and policy checks across multiple dimensions

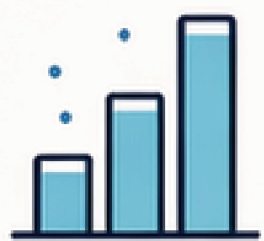


Outcome Classification

Route to accept (forward) or reject (blacklist) upstream of secure aggregation



Scenario 1: Honest Update



- ✓ proof valid
- ✓ chain consistency
- ✓ accepted & forwarded



Scenario 2: Teleportation (A1)



- ✗ per-step bound violated
- ✗ Groth16 fails
- ✗ rejected



Scenario 3: Replay (A5)



- ✗ stale predecessor commitment
- ✗ chain-state check fails
- ✗ rejected & blacklisted



Scenario 4: Identity Swap (A2)



- ✗ secret binding fails
- ✗ public-signal / proof binding fails
- ✗ rejected & blacklisted



Scenario 5: Gradual Drift (A3)



- ✗ ADWC policy cap exceeded
- ✗ rejected by ADWC



Benign updates pass and are forwarded.



Invalid cross-round behaviors are rejected and blacklisted upstream of secure aggregation.